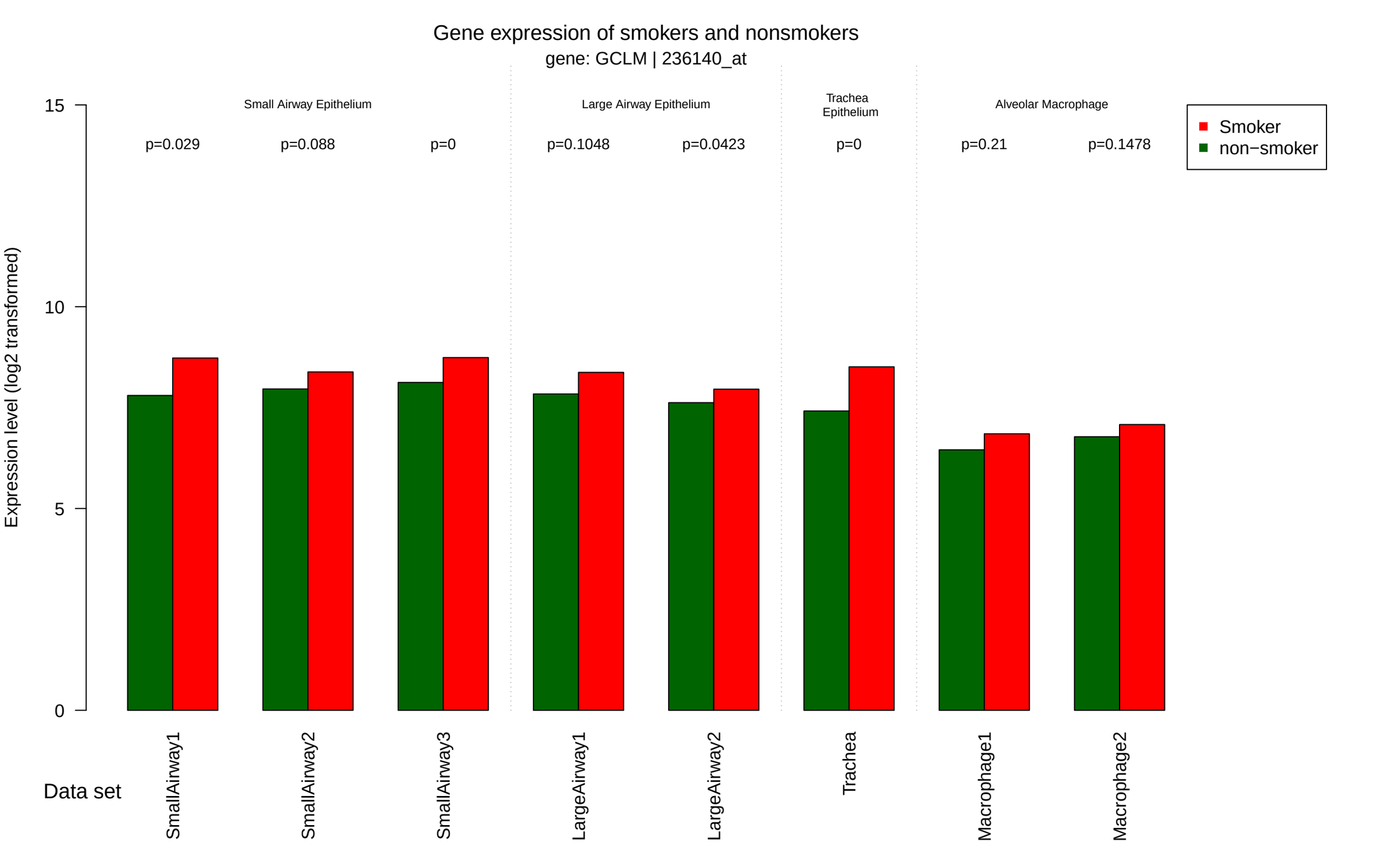
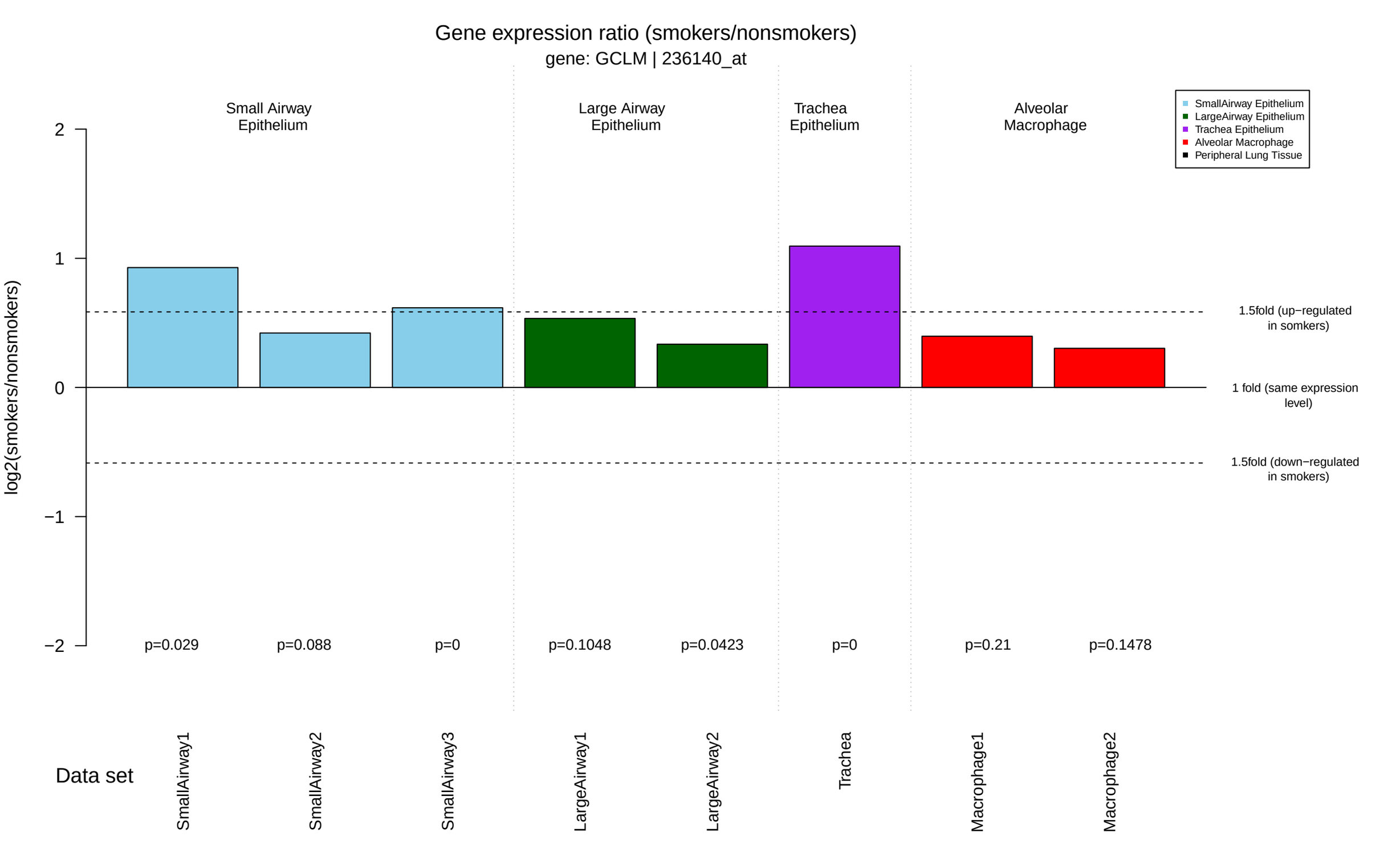
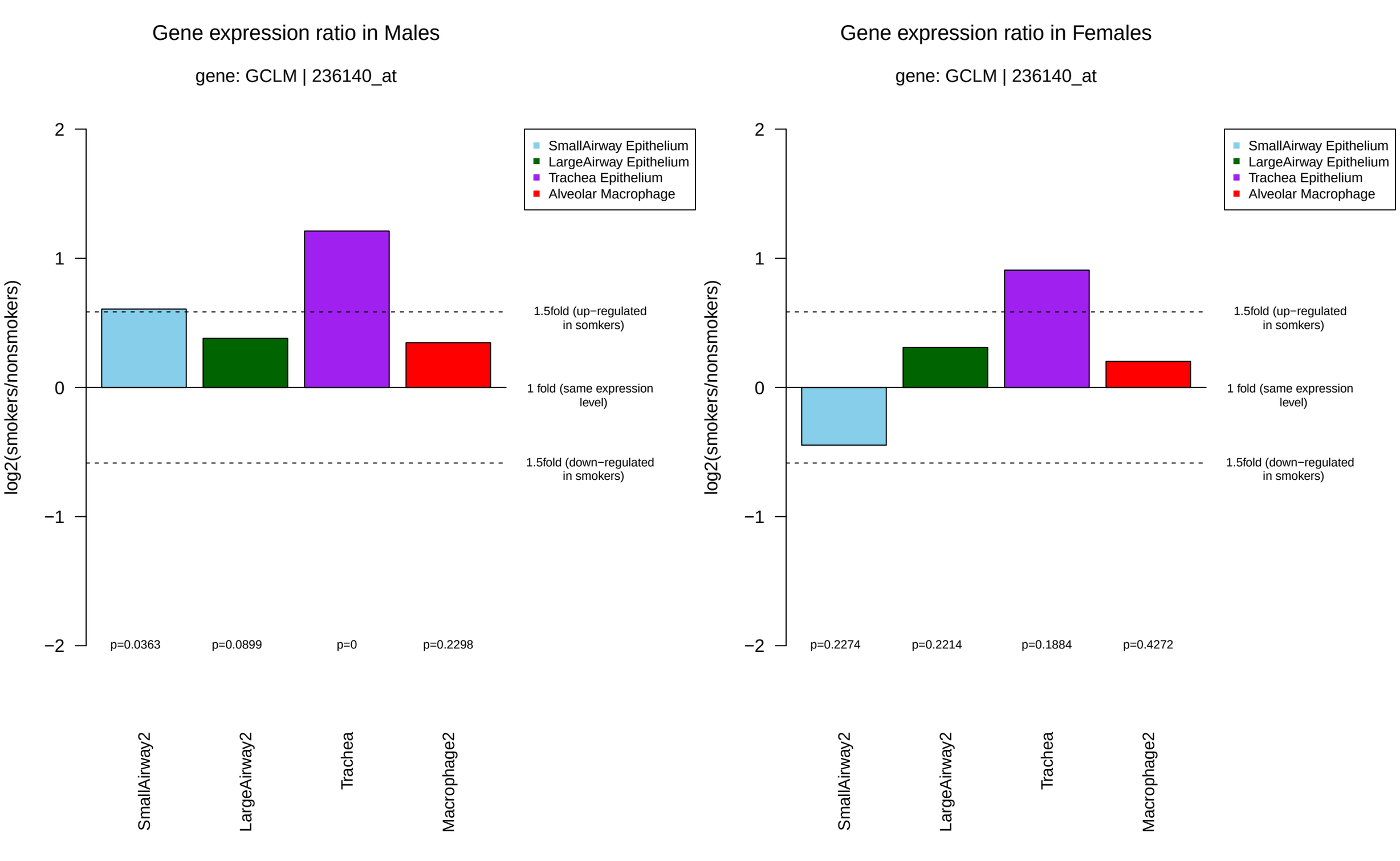
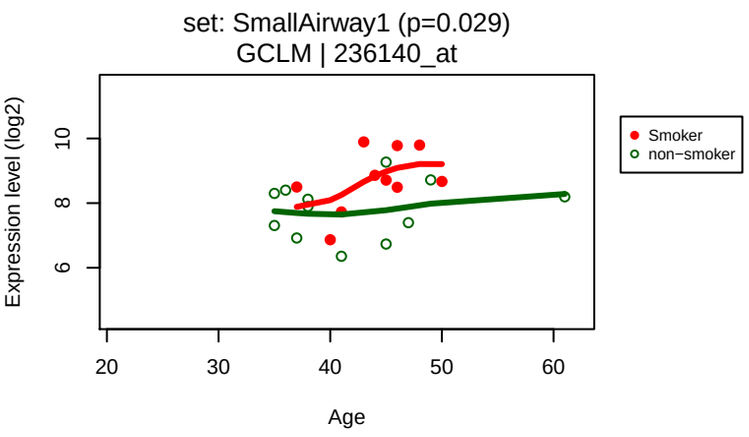
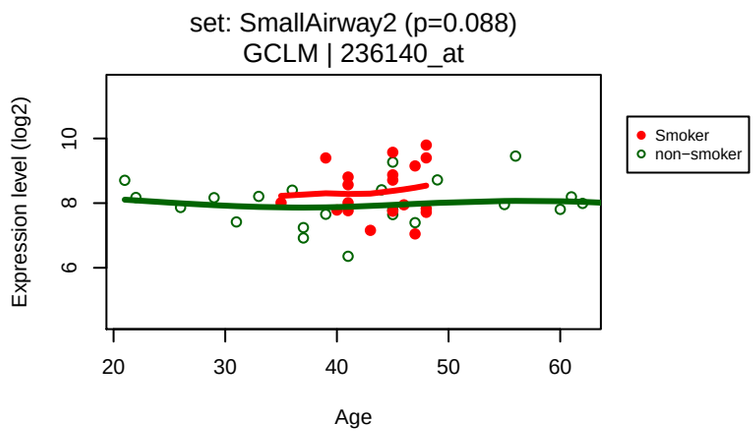
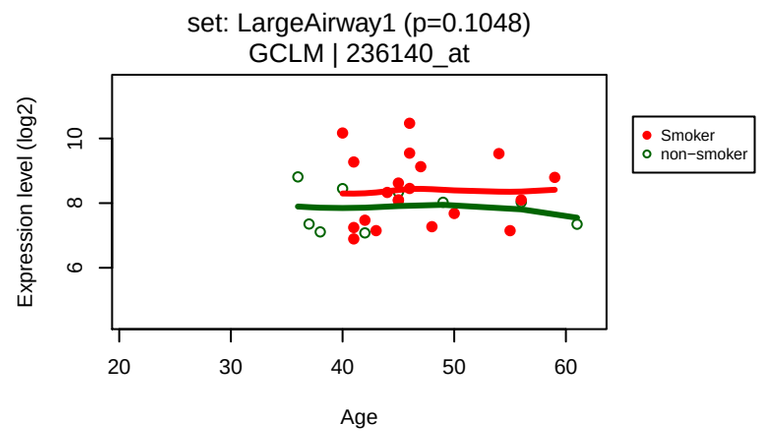

Gene expression of smokers and nonsmokers
gene: GCLM | 236140_at
Small Airway Epithelium
Large Airway Epithelium
Trachea Epithelium
Alveolar Macrophage
p=0.029
p=0.088
p=0
p=0.1048
p=0.0423
p=0
p=0.21
p=0.1478
Smoker
non−smoker
Expression level (log2 transformed)
15
10
5
0
Data set
SmallAirway1
SmallAirway2
SmallAirway3
LargeAirway1
LargeAirway2
Trachea
Macrophage1
Macrophage2

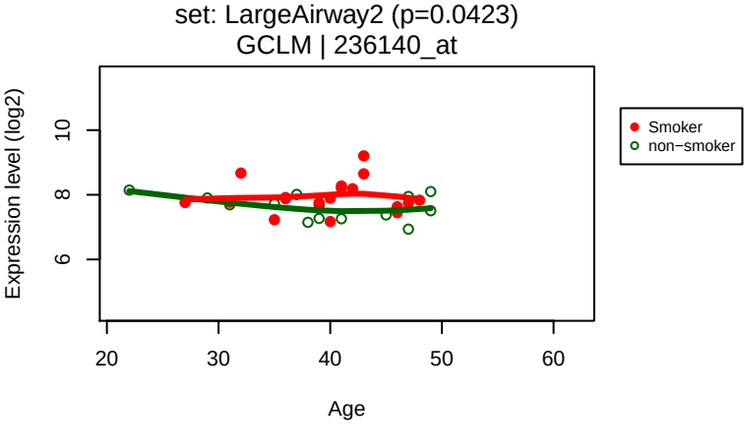

Gene expression ratio (smokers/nonsmokers)
gene: GCLM | 236140_at
Small Airway Epithelium
Large Airway Epithelium
Trachea Epithelium
Alveolar Macrophage
SmallAirway Epithelium
LargeAirway Epithelium
Trachea Epithelium
Alveolar Macrophage
Peripheral Lung Tissue
log2(smokers/nonsmokers)
2
1
0
−1
−2
1.5fold (up−regulated in somkers)
1 fold (same expression level)
1.5fold (down−regulated in smokers)
p=0.029
p=0.088
p=0
p=0.1048
p=0.0423
p=0
p=0.21
p=0.1478
Data set
SmallAirway1
SmallAirway2
SmallAirway3
LargeAirway1
LargeAirway2
Trachea
Macrophage1
Macrophage2

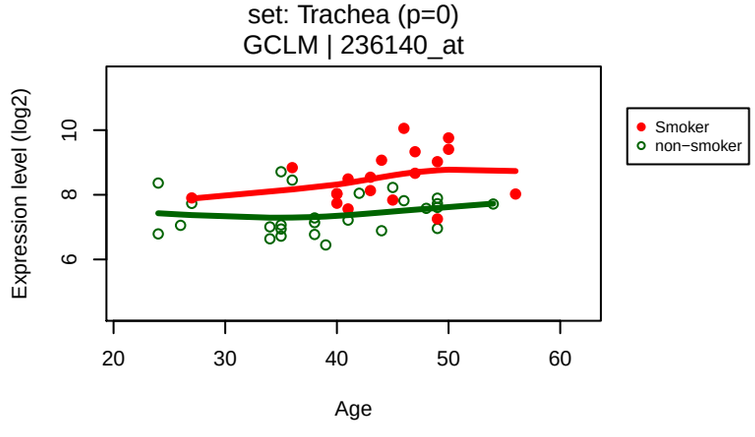

Gene expression ratio in Males
gene: GCLM | 236140_at
log2(smokers/nonsmokers)
2
1
0
−1
−2
SmallAirway Epithelium
LargeAirway Epithelium
Trachea Epithelium
Alveolar Macrophage
1.5fold (up−regulated in somkers)
1 fold (same expression level)
1.5fold (down−regulated in smokers)
p=0.0363
p=0.0899
p=0
p=0.2298
SmallAirway2
LargeAirway2
Trachea
Macrophage2
Gene expression ratio in Females
gene: GCLM | 236140_at
log2(smokers/nonsmokers)
2
1
0
−1
−2
SmallAirway Epithelium
LargeAirway Epithelium
Trachea Epithelium
Alveolar Macrophage
1.5fold (up−regulated in somkers)
1 fold (same expression level)
1.5fold (down−regulated in smokers)
p=0.2274
p=0.2214
p=0.1884
p=0.4272
SmallAirway2
LargeAirway2
Trachea
Macrophage2

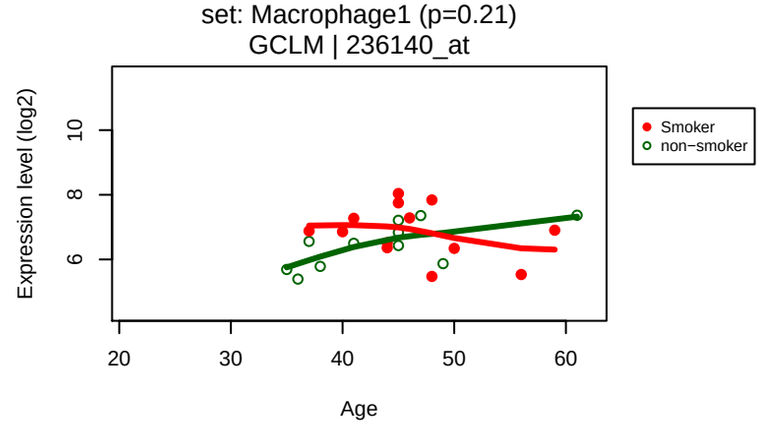

set: SmallAirway1 (p=0.029)
GCLM | 236140_at
Expression level (log2)
Age
Smoker
non−smoker

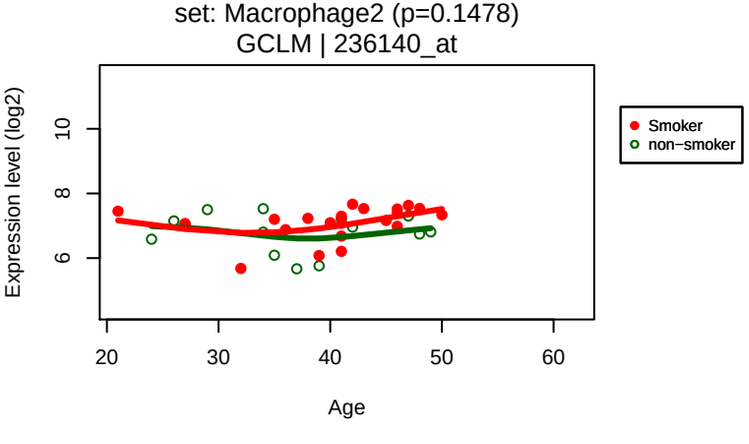

set: SmallAirway2 (p=0.088)
GCLM | 236140_at
Expression level (log2)
Age
Smoker
non−smoker

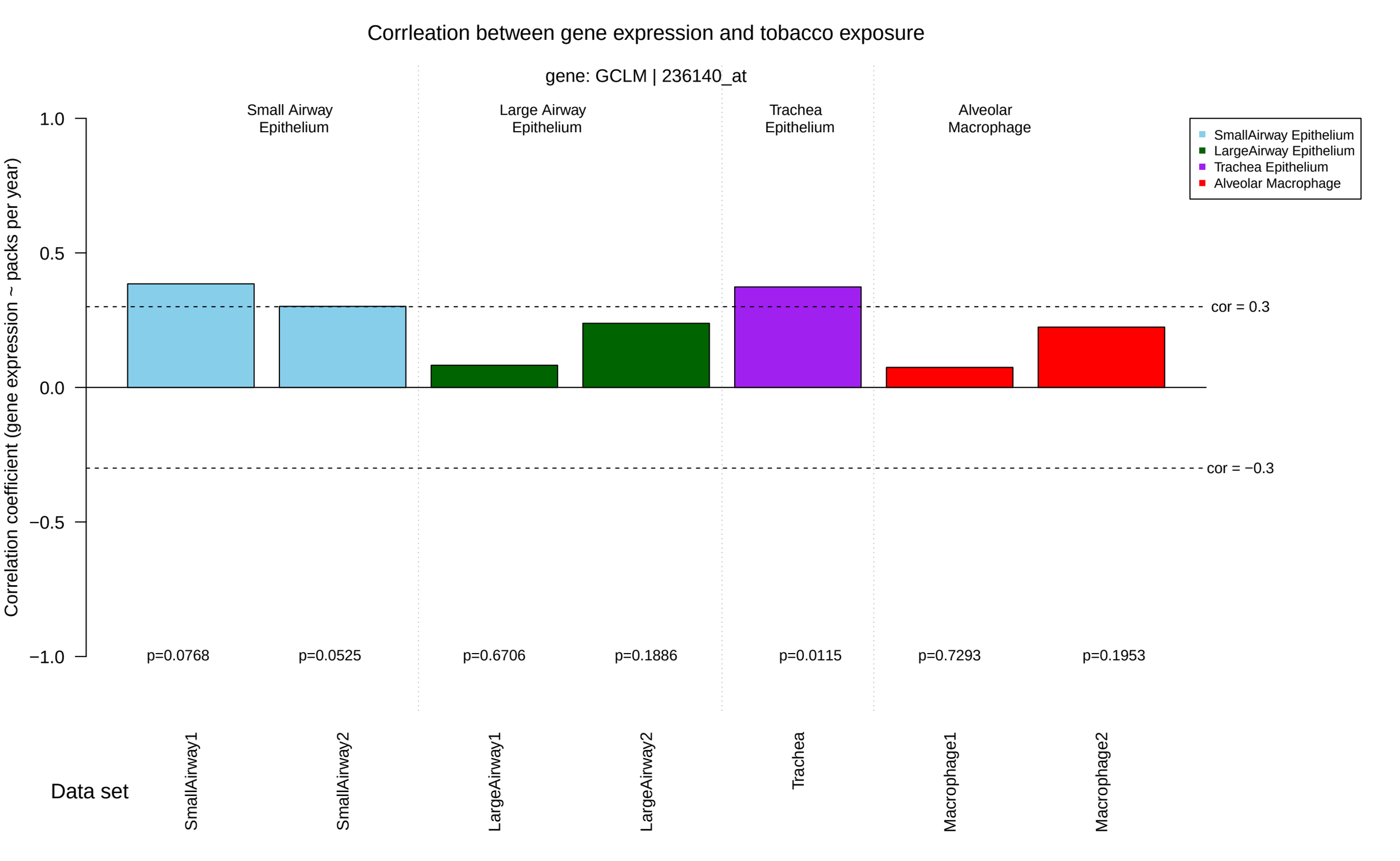

set: LargeAirway1 (p=0.1048)
GCLM | 236140_at
Expression level (log2)
Age
Smoker
non−smoker


set: LargeAirway2 (p=0.0423)
GCLM | 236140_at
Expression level (log2)
Age
Smoker
non−smoker


set: Trachea (p=0)
GCLM | 236140_at
Expression level (log2)
Age
Smoker
non−smoker


set: Macrophage1 (p=0.21)
GCLM | 236140_at
Expression level (log2)
Age
Smoker
non−smoker


set: Macrophage2 (p=0.1478)
GCLM | 236140_at
Expression level (log2)
Age
Smoker
non−smoker


Corrleation between gene expression and tobacco exposure
gene: GCLM | 236140_at
Small Airway Epithelium
Large Airway Epithelium
Trachea Epithelium
Alveolar Macrophage
SmallAirway Epithelium
LargeAirway Epithelium
Trachea Epithelium
Alveolar Macrophage
Correlation coefficient (gene expression ~ packs per year)
1.0
0.5
0.0
−0.5
−1.0
cor = 0.3
cor = −0.3
p=0.0768
p=0.0525
p=0.6706
p=0.1886
p=0.0115
p=0.7293
p=0.1953
Data set
SmallAirway1
SmallAirway2
LargeAirway1
LargeAirway2
Trachea
Macrophage1
Macrophage2